

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: SUB-STRAND 1.6 PLANT GROWTH AND DEVELOPMENT: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

In this Final Explanation you will use everything you have learned in Lessons 1 to 8 to answer the driving question: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? Remember the Kisumu farmer who planted maize and beans in the same bed, on the same day, in the same soil, with equal watering. Answer each section in the blank space under the prompt. Write in full sentences, use the correct science words (for example dormancy, epigeal, hypogeal, apical meristem, vascular cambium, auxin, abscisic acid), and refer to the evidence you saw in class. You may draw and label small diagrams where a prompt asks for one. Each answer should link back to the Kisumu farmer's garden.

Section 1: Why Did Some Stored Seeds Refuse to Germinate?

The Kisumu farmer found that some seeds from a packet stored for two seasons did not germinate, even when given water and warmth.

(a) Explain what dormancy is and say whether a dormant seed is dead or alive. (b) Describe TWO causes of dormancy that could explain why the stored seeds failed to germinate even though they had water and warmth. (c) Describe TWO ways the farmer could break dormancy before planting, linking each to a practice used by farmers in Kisumu. (d) Name the three external conditions a seed needs for germination, and state what each one does inside the seed.

(a) Dormancy is a state in which a living seed does not germinate because germination is blocked. A dormant seed is alive, not dead. Dormancy is a survival strategy that kept the stored seeds alive across two seasons.

(b) Any two of the following. A hard, impermeable seed coat can hinder water and air from entering the seed, even when water is available. Chemical inhibitors such as abscisic acid, which may have built up during long dry storage, can hold the embryo back from growing. An immature embryo or unfavourable environmental conditions are two further causes taught in class.

(c) Any two of the following. Scarification means nicking or scratching the seed coat so that water can enter; bean farmers around Kisumu sometimes nick seeds before planting. Soaking softens the seed coat and starts water uptake; bean farmers around Kisumu also soak seeds before planting. Stratification (a cold treatment) is a third method taught in class.

(d) Water triggers imbibition, so the seed swells and its enzymes become active. Oxygen powers aerobic respiration, which releases energy for cell growth. An optimum temperature lets the enzymes work efficiently. All three must be present together. Removing even one prevents or delays germination, as our experiment showed with the cups that had no water, no air or a cold location. In the farmer's stored seeds, however, water and warmth were present, so the failure to germinate points to dormancy such as a hardened seed coat or chemical inhibitors.

Section 2: Why Do Maize and Bean Seedlings Emerge Differently?

Within a week the Kisumu farmer saw bean seedlings push their seed leaves above the soil in an arch, while maize sent up a single green spike and the seed stayed underground.

(a) Name the type of germination shown by beans and the type shown by maize. (b) Explain, using the words hypocotyl, epicotyl, cotyledon and plumule, how each type of seedling gets above the soil. (c) Explain one advantage of each strategy for the young seedling. (d) The bean and maize seeds were planted in the same soil with the same water. What does this tell you about the cause of their different emergence, and which part of the seedling's structure differs between the two?

- (a) Beans show epigeal germination. Maize shows hypogeal germination.
- (b) In beans the hypocotyl elongates and forms an arch that pulls the cotyledons above the soil, and the plumule follows. In maize the epicotyl elongates and pushes the plumule upward as a straight spike, while the cotyledon stays underground.
- (c) In beans the cotyledons rise into the light, where they can begin photosynthesis. In maize the energy-rich cotyledon stays safely underground as a food reserve while the plumule grows. Both strategies establish the seedling successfully.
- (d) Because soil, water and planting day were the same, the difference cannot be caused by the garden conditions. It comes from the seeds themselves: the two species differ in which part elongates (the hypocotyl in beans, the epicotyl in maize) and so in where the cotyledons end up, above or below the soil.

Section 3: Getting Taller and Getting Thicker

Next to the bean plants in the farmer's garden are mango and avocado trees planted years earlier. They have thick, woody trunks, while the bean plants stay thin.

(a) Explain how primary growth and secondary growth are different, naming the tissue responsible for each. (b) Describe what the vascular cambium produces each growing season and how this forms the rings seen in the mango trunk photograph. (c) Explain why bean plants,

- (a) Primary growth increases the length of roots and shoots. It happens through cell division in the apical meristems at the tips. Secondary growth increases the girth (thickness) of stems and roots. It happens through cell division in the vascular cambium, a ring of lateral meristem between the xylem and phloem.
- (b) Each growing season the vascular cambium produces new xylem inward and new phloem outward. Each season's new wood forms a layer, seen as one annual ring in the trunk cross-section.
- (c) These herbaceous plants have little or no vascular cambium activity. They grow mainly by apical meristem activity, which makes them longer but not thick and woody.
- (d) Counting the rings shows the age of the tree, because each ring is one growing season. The dense, strong timber of Kenyan trees is the accumulated product of many years of secondary growth.

sukuma wiki and maize do not develop thick woody trunks. (d) What can a person find out by counting the rings in a mango trunk?	
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Section 4: Chemical Control and the Environment

(a) Match each hormone to ONE job it does in the plant: auxins, gibberellins, cytokinins, abscisic acid, ethylene. (b) Sukuma wiki grown near a window bends toward the light. Explain how auxin causes this. (c) In a Kisumu market, mangoes stored in a crate on their own ripen slowly, but mangoes stored in a crate with ripe bananas ripen much faster. Explain why. (d) Choose ONE of the regional case studies (tea in Kericho, maize in the Rift Valley, or mangroves in Mombasa). Explain how water, temperature, light or nutrients affect how plants grow there, and how these factors work together with hormones or growth type.	(a) Auxins control phototropic bending and apical dominance by causing unequal cell elongation. Gibberellins promote stem elongation and help break seed dormancy. Cytokinins stimulate cell division and lateral bud growth. Absciscic acid maintains seed dormancy and closes stomata during drought. Ethylene triggers fruit ripening and leaf drop. (b) Auxin moves to the shaded side of the stem. The cells on the shaded side elongate more than the cells on the lit side, so the stem bends toward the light. (c) Ripe fruits release ethylene, a plant hormone that triggers ripening. Ethylene released by the bananas triggers ripening in the nearby mangoes, so they ripen faster together. (d) Example, Kericho tea: cool temperatures, high rainfall and shade give steady water and moderate temperatures, so enzymes work well and the plants grow and branch continuously. These factors do not act alone. They interact with hormone action and with the type of growth inside the plant, so the environment shapes how fast the plant grows and what form it takes. (Accept a correct answer using the Rift Valley maize or Mombasa mangrove case study, naming at least one environmental factor and linking it to growth.)
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Section 5: Explaining a New Problem and the Whole Phenomenon

A farmer in Nakuru plants bean seeds in a shaded corner of her plot. After two weeks the seedlings are much taller than seedlings in the sunny part of the plot, but their stems are thin and weak, their leaves are a washed-out colour, and they have few side branches. (a) Name the process that has caused these differences and	(a) The seedlings are etiolated. The environmental factor is low light. (b) In low light, auxin and gibberellin activity increases the elongation of the stem cells, so the seedlings put their growth into length and become long and thin. Apical dominance, controlled by auxin from the apical bud, suppresses the growth of lateral buds, so there are few side branches. (c) Any two sensible actions with a reason. She could move the seedlings, or replant them, in a sunnier place, which removes the low-light cause of etiolation. She could also thin out or remove shading plants or objects, which lets more light reach the seedlings for the same reason. Breaking apical dominance by removing the apical bud would only encourage side branches; it would not fix the low light, so it does not solve the cause of etiolation. (d) Seeds from the same garden differ because maize and beans differ in germination type, so beans emerge by the epigeal pattern and maize by the hypogeal pattern. Some seeds do not germinate because dormancy, held by abscisic acid or a hard seed coat, blocks them even when water and warmth are present. After germination, primary growth at apical meristems increases
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the environmental factor involved. (b) Use auxin, gibberellin and apical dominance to explain why the shaded seedlings are so long and thin with little branching. (c) Suggest TWO actions the farmer could take and explain how each works. (d) Now answer the driving question in 4 to 6 sentences: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? Use ideas from dormancy, germination type, growth type, hormones and the environment.	length, and in trees secondary growth at the vascular cambium increases girth and builds woody trunks. Hormones such as auxins, gibberellins, cytokinins, abscisic acid and ethylene control each stage. Water, temperature, light and nutrients interact with these hormones, so the same garden produces different plant forms.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of science ideas (dormancy, germination, growth, hormones)	Uses all key ideas accurately and in detail, including correct terms such as epigeal, hypogeal, apical meristem, vascular cambium and the roles of each hormone.	Uses most key ideas correctly with only minor errors or missing detail.	Gives few ideas, or uses terms incorrectly, or confuses growth types, hormones or dormancy.
Linking to the Kisumu phenomenon and evidence	Clearly connects every answer to the Kisumu farmer's garden and to class evidence such as the germination containers, mango trunk rings and the germination experiment cups.	Connects most answers to the phenomenon or class evidence.	Gives general answers with little or no link to the phenomenon or evidence.
Explaining cause and effect	Explains how and why, step by step, for example how auxin redistribution causes bending, how increased elongation causes etiolation, and how cambium makes rings.	Gives correct causes but some steps of the reasoning are missing.	States facts without explaining why or how, or the reasoning is incorrect.
Applying ideas to a new problem (Nakuru and regional cases)	Correctly diagnoses etiolation from low light, gives solutions that address the cause, and integrates environment with hormones and growth.	Diagnoses the problem and gives a sensible solution with partial explanation.	Does not identify the cause or gives solutions without scientific reasons.

Clarity and organisation	Answers are complete, well organised and use accurate labelled diagrams or sentences where asked.	Answers are mostly clear and organised, with small gaps.	Answers are unclear, incomplete or hard to follow.
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